Week : 9

الأسبوع : من - إلى

Subject/:الموضوع Chemistry (IB DP) **Unit Title : عنوان الوحدة Structure 1.5 – Ideal gases** **Grade: 11 (DP1)**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 **S1.5.3 Molar volume of gases**	**Learning target (DOK 2):** I can explain the relationship between temperature, pressure and volume for a fixed mass of an ideal gas, and analyse graphs relating these variables.	**UDL strategies:** 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	IB Chemistry Interactive: PhET Gas Properties simulation – see how pressure, volume and temperature change for a fixed amount of gas. Class Teams (Structure 1.5 folder): lesson slides and graph practice. IB data booklet: molar volume at STP is given.
Lesson 2 **الحصة الثانية** **S1.5.4 Ideal gas equation**	**Learning target (DOK 2):** I can solve problems using the ideal gas equation PV = nRT and the combined gas law, using SI units only.	**UDL strategies:** 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	IB Chemistry Interactive: PhET Gas Properties simulation – test how changing one variable affects the others. Class Teams (Structure 1.5 folder): lesson slides and ideal gas practice questions. IB data booklet: R, the ideal gas equation and the combined gas law are given.
Lesson 3 **الحصة الثالثة** **Structure 1 formative assessment**	**Learning target (DOK 2):** I can apply my understanding of Structure 1 (particles, the mole and gases) to solve problems and explain my reasoning in a formative assessment.	**UDL strategies:** 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays. 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	IB Chemistry Class Teams (Structure 1 folder): revision resources and practice questions for Structure 1. At home: revise your Structure 1 notes in Teams before the formative.
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Bring a calculator, Chemistry notebook, ruler, pencil, and coloured pens. Participate actively, complete all classwork, correct misconceptions, and upload required work to Toddle by the stated deadline		

Week الأسبوع

Technology Integration دمج التكنولوجيا	Videos posted on Teams; PhET Gas Properties simulation; mini whiteboards; scientific calculators; digital IB data booklet; Teams for the formative assessment.
Cross-curricular links الروابط بين المواد الدراسية	**Mathematics:** rearranging equations, sketching graphs and unit conversions. **Physics:** pressure and temperature in kelvin.
External Assessment Integration دمج التقييمات الخارجية	IB DP Chemistry Papers 1 and 2: ideal gas calculations in SI units, sketching and interpreting graphs, and data-booklet use, with the command terms calculate, sketch, determine and explain.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking:** analysing graphs and justifying answers. **Communication:** showing clear working with SI units. **Self-management:** reflecting on formative results and setting next steps.
Standards المعايير	• **S1.5.3** The molar volume of an ideal gas is a constant at a specific temperature and pressure. • **S1.5.4** The relationship between the pressure, volume, temperature and amount of an ideal gas is shown in the ideal gas equation PV = nRT and the combined gas law $P_1V_1/T_1 = P_2V_2/T_2$.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Nature of science, Tools 2 and 3, Reactivity 2.2:** Graphs can be presented as sketches or as accurately plotted data points. What are the advantages and limitations of each representation? **Tool 1, Inquiry 2:** How can the ideal gas law be used to calculate the molar mass of a gas from experimental data? **Lesson 3:** How can a formative assessment show what we know and what we still need to learn?
Success Criteria معايير النجاح **(Measurable)**	● **Must (DOK 1–2):** state that the molar volume of an ideal gas is constant at a given temperature and pressure, and state PV = nRT with SI units. ■ **Should (DOK 2–3):** use PV = nRT and the combined gas law to solve problems, and sketch graphs relating P, V and T. ▲ **Could (DOK 3–4):** evaluate sketched against plotted graphs and calculate the molar mass of a gas from experimental data. **Evidence:** hinge questions, exit tickets and the Structure 1 formative.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Global Understanding التفاهم العالمي Heritage التراث المجال Area: Innovation and Infrastructure التطبيق Implementation: Students link gas laws to UAE contexts: tyre pressure in summer heat and the storage and transport of natural gas.